DATA ANALYSIS MINI PROJECT · DEPI, ITC ONLINE GROUP 3, TEAM 3

Olist Brazilian E-Commerce: from raw data to dashboard

Dataset discovery, SQL database, Python cleaning, Power BI measures and a four-page dashboard covering Jan 2017 to Aug 2018

Objective and why this dataset

Objective

Understand sales, customers and delivery performance of a Brazilian online marketplace.

The dataset

Brazilian E-Commerce Public Dataset by Olist, found on Kaggle by Mohamed Sokarno.

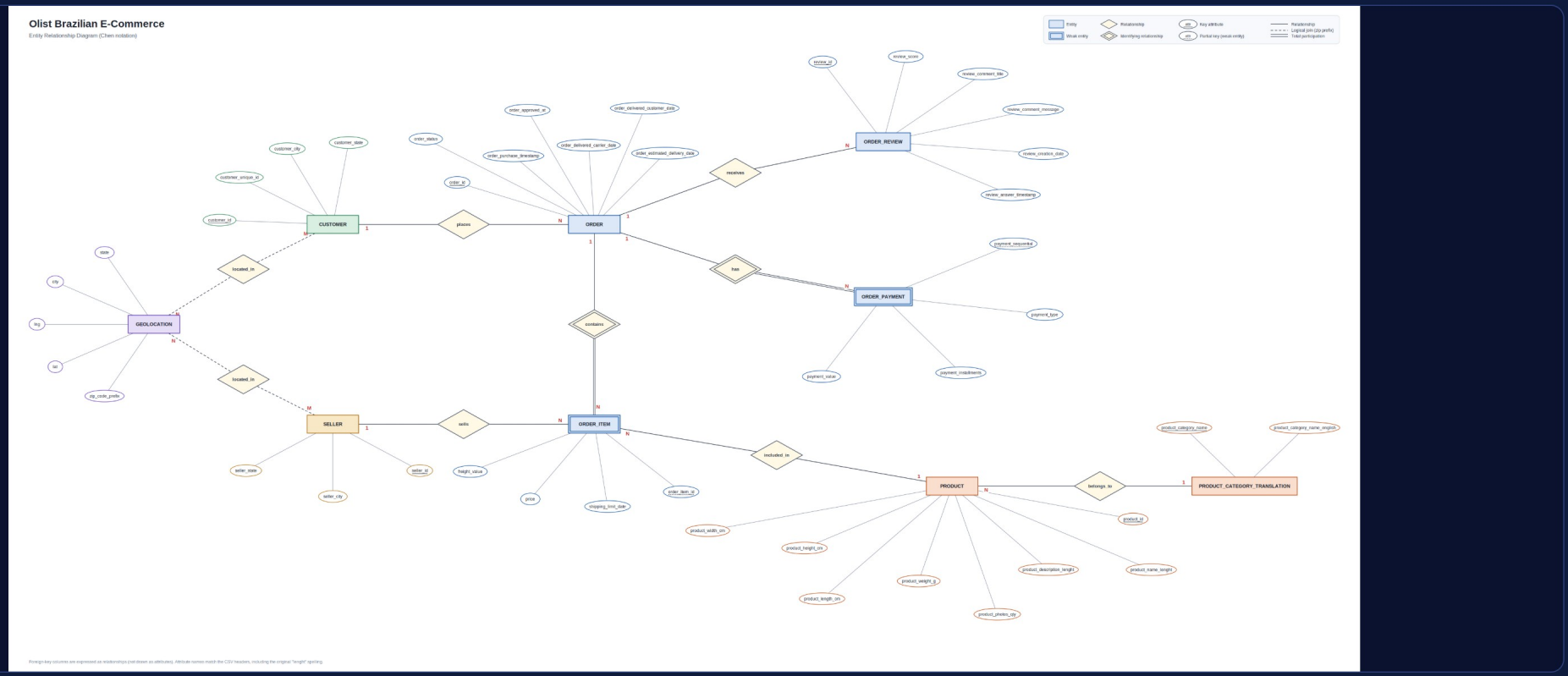
Why it fits

A relational structure of customers, orders, items, products and more, suited to SQL, cleaning and BI work.

THE PROJECT JOURNEY

1. Discover
2. Clean
3. Model in SQL
4. Measure
5. Dashboard

The original dataset and its structure



Source ERD (Chen notation) by Mahmoud Al-Nabawi: nine entities including customers, orders, items, payments, reviews, sellers, products and geolocation.

Data cleaning and quality checks

Clean the cells

Removed empty rows and duplicates; standardised column names and text; handled missing values; corrected data types.

Check unusual values

Looked for outliers and unexpected values in each column.

Check relationships

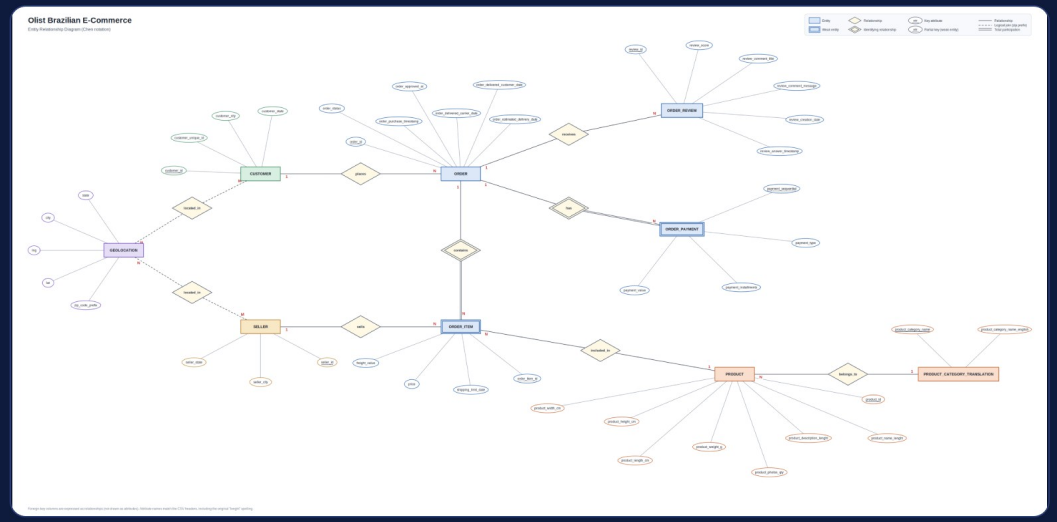
Orders reference existing customers; order items reference existing products; dates stay consistent across tables.

Flag, do not delete

Suspicious records were flagged for review instead of being removed automatically.

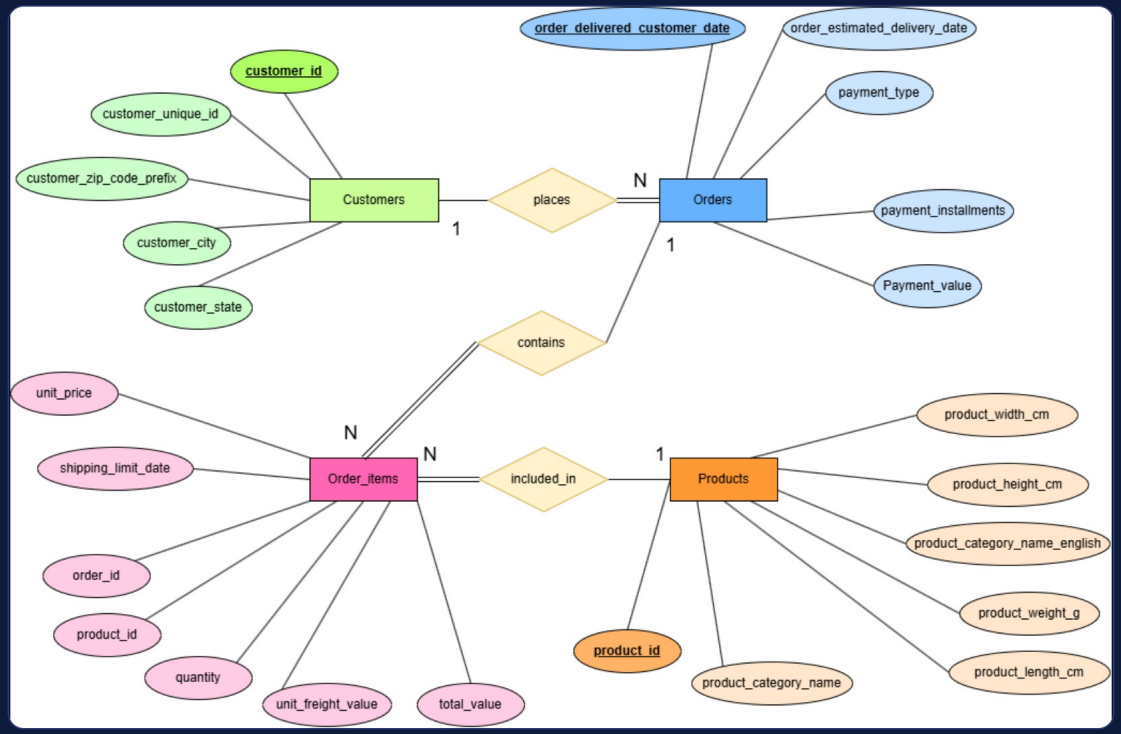
Lesson: data quality is also about relationships between tables, not only clean cells. Python notebook by Mohamed El-Gml.

From source ERD to the final four-table schema



Before: original source with nine entities.

Key relationships: one Customer places many Orders; one Order contains many Order items; one Product is included in many Order items.



After: Customers, Orders, Order_items, Products.

SQL implementation



```sql
-- Creating Customers
CREATE TABLE Customers
(
    customer_id VARCHAR(32) NOT NULL,
    customer_unique_id VARCHAR(32) NOT NULL,
    customer_zip_code_prefix VARCHAR(5) NOT NULL,
    customer_city VARCHAR(100) NOT NULL,
    customer_state VARCHAR(2) NOT NULL,

    geolocation_zip_code_prefix VARCHAR(5) NOT NULL,
    geolocation_lat DECIMAL(10, 8),
    geolocation_lng DECIMAL(11, 8),
    geolocation_city VARCHAR(100),
    geolocation_state VARCHAR(2),

    CONSTRAINT PK_Customers
        PRIMARY KEY (customer_id)
);
GO

select * from Customers


-- Filling Customers

INSERT INTO Customers
(
    customer_id,
    customer_unique_id,
    customer_zip_code_prefix,
    customer_city,
```

Schema and tables

Database, schema and tables created in SQL, e.g. Customers with customer_id as primary key.

Loading

Data loaded with INSERT statements and joins into the four target tables.

Encoding fix

Text encoding issues were investigated and corrected by converting to UTF-8.

Open item: the payment_installments Boolean note is not confirmed by the final ERD.

Power BI measures and the Calendar table

Sales and revenue

Sales, freight, revenue, average order value, units, products.

Orders and customers

Delivered orders, unique and repeat customers (based on customer_unique_id).

Time intelligence

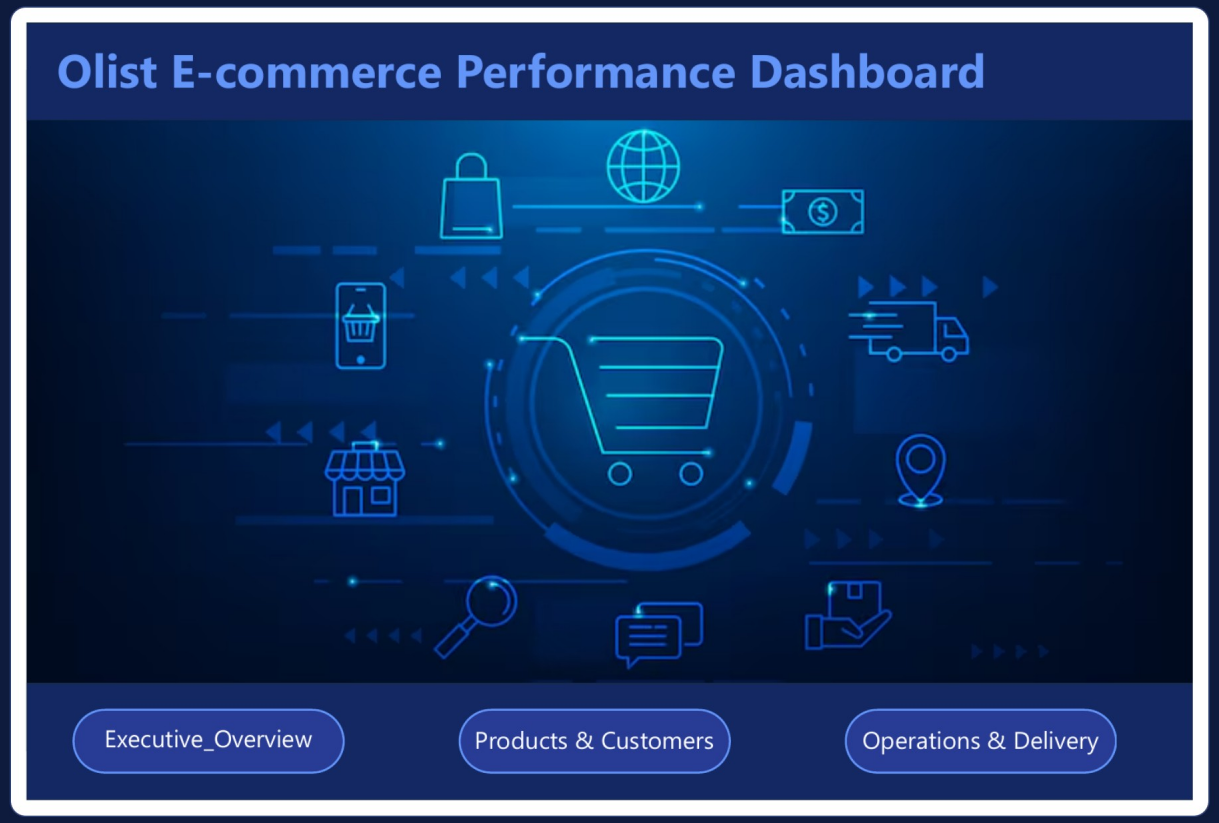
Calendar table plus YTD and month-over-month comparisons; guarded MoM hides results when the prior-month base is under 1,000.

Cancellations and delivery

Cancellations, late and on-time delivery logic.

Built by Mahmoud Saqr, as described in the project notes. Final DAX definitions still to be validated.

Dashboard: four pages, Jan 2017 to Aug 2018



Home

Navigation buttons to the three report pages.

Executive Overview

Revenue, sales, freight, orders, trend, geography, payments.

Products & Customers

Units, categories, repeat rate, monthly sales, YTD and MoM.

Operations & Delivery

Delivery days, late rate, location comparison, decomposition tree.

Executive Overview



15.4M

Total revenue (13.2M product sales + 2.2M freight)

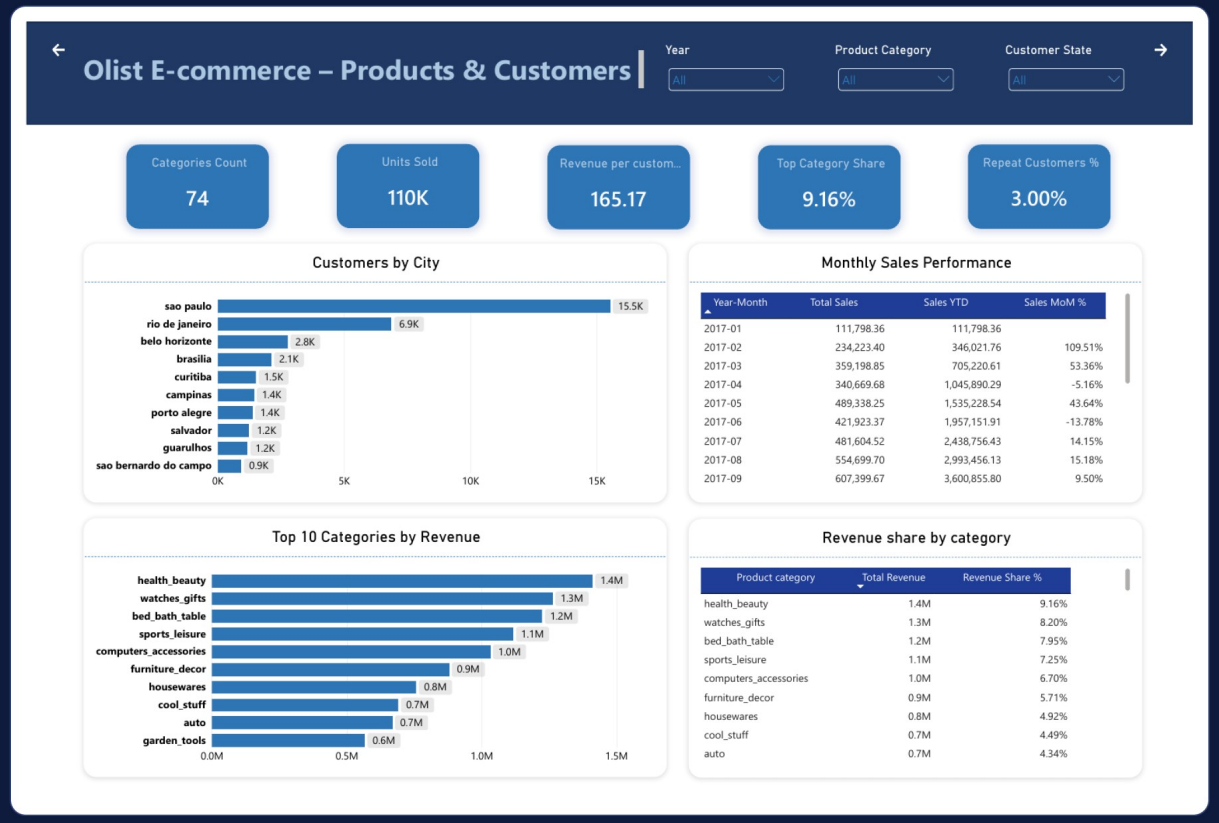
96.5K

Delivered orders; 93.4K unique customers

137

Average order value; 78.4% of revenue by credit card

Products & Customers



110K

Units sold across 74 categories

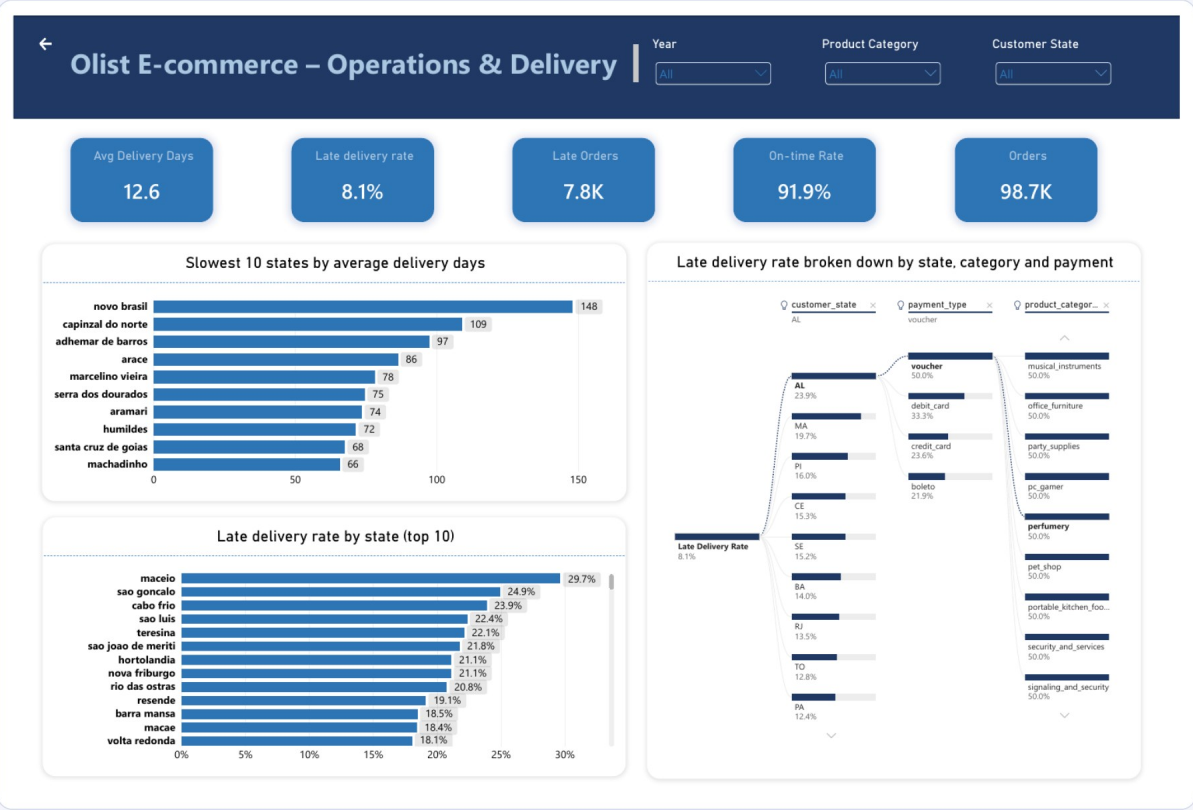
9.16%

Top category share: health_beauty, then watches_gifts and bed_bath_table

3.00%

Repeat customers; revenue per customer 165.17

Operations & Delivery



12.6

Average delivery days

8.1%

Late-delivery rate (7.8K late orders); on-time rate 91.9%

Caution: small samples (e.g. categories at 50% late) can mislead.

Main findings and possible implications

15.4M
Total revenue

38%
of revenue from São Paulo state (5.8M)

39%
of revenue from the top five categories

3%
repeat customers

Delivery: 12.6 days on average, 8.1% late. Late rates are higher in some northeastern states.

Hypotheses to investigate, not proven causes: low repeat purchase suggests a retention opportunity; regional late rates suggest reviewing delivery estimates and carriers.

Limitations, trade-offs and lessons

Source scope unverified

Final database has four tables; the claim that eight of nine source tables were used is not confirmed.

Late is not caused

Delivery after the estimate does not show whether carriers or estimates are at fault.

Small samples mislead

Tiny cities or categories skew averages and late rates; use a minimum sample threshold.

Labels and data gaps

Some location charts label cities as states; identifiers are used instead of names; currency is not stated.

Credits: project contributors

Mohamed Sokarno

Dataset discovery, database creation, SQL schema and loading, UTF-8 fix, project review, repository organisation, coordination, documentation and narrative.

Mahmoud Al-Nabawi

Source-dataset ERD and contribution to the data dictionary.

Mohamed El-Gml

Python data cleaning, preparation and notebook.

Mahmoud Saqr

Power BI measures and Calendar table.

Mostafa Salem

Four-page Power BI dashboard.

Leadership and program acknowledgements

Mohamed Sokarno
Team Leader.

Eman Ghanem
DEPI mentor and lecturer.

Esraa Yahia
ITC_ONL5_Dat3_G3 operator.

Mahmoud Raslan
ITC CEO.

Program acknowledgements recognise support and guidance; project components were implemented by the contributors listed on the previous slide.